Fiscal Year:	FY 2021	**Task Last Updated:**	FY 05/25/2021
PI Name:	Goree, John Ph.D.		
Project Title:	Nonlinear Wave Experiments in Dusty Plasmas		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:	National Science Foundation	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	john-goree@uiowa.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	319-335-1843
Organization Name:	University of Iowa		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	210 Van Allen Hall		
PI Web Page:			
City:	Iowa City	**State:**	IA
Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	06/14/2017	**End Date:**	12/13/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
Contact Email:	john.l.callas@jpl.nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/13/2021 per U. Israelsson/JPL (Ed., 5/25/21) NOTE: Change in start and end dates to 6/14/2017-6/13/2021 (from 4/15/2017-4/15/2021) per I. Hahn/JPL and PI (Ed., 1/19/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Liu, Bin Ph.D. (University of Iowa)		
Grant/Contract No.:	JPL 1573629		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

A flight experiment will be performed and analyzed in a study of nonlinear synchronization of waves. Synchronization is a cooperative phenomenon for oscillators or waves. A signature of synchronization is a frequency that shifts when an external modulation is applied. Compared to point-like oscillators, waves are more complicated due to their spatio-temporal behavior and dispersion properties. An understanding of synchronization is lacking for waves of all kinds, including those in dusty plasmas.
The experiment will use dusty plasmas, also called "complex plasmas." These are mixtures of small particles of solid matter, electrons, ions, and neutral gas. The small solid particles obtain large electric charges, so that they exhibit collective behavior. They are electrically confined, so that they do not contact any boundaries.

Microgravity enables this experiment by suppressing sedimentation, thereby allowing the waves to propagate distances of many wavelengths, with relatively uniform background conditions. This experiment is based on previous ground-based experiments, where waves propagated only a few wavelengths.

Our flight experiment will be performed using “Plasma Kristall 4” (PK-4), which is a facility in the Columbus Module of the International Space Station (ISS), built by the European Space Agency (ESA).

SCIENTIFIC/TECHNICAL MERITS

The main scientific topic is the synchronization of a dust acoustic wave (DAW). A DAW is a low-frequency compressional wave, which involves charge-separation from electrons and ions. The DAW is self-excited by an ion flow, and it grows naturally to a large amplitude so that nonlinear effects occur readily. Synchronization experiments are performed by applying electrical waveforms to an electrode to modulate the dusty plasma sinusoidally. The diagnostic for detecting the wave motion is video imaging using particle observation cameras. Image data will be analyzed to prepare space-time diagrams, which provide a spatial and temporal description of the wave's amplitude as it propagates. These space-time diagrams are the foundation for further analysis methods.

In support of our experiment, a theoretical effort will be undertaken at Iowa. The goal is to contribute to the development of a theory for synchronization of propagating waves, to compare to the experiment. The theoretical effort at Iowa will focus on numerical simulations using a hybrid model. Molecular dynamics simulation will also be performed. The main theoretical effort, which will center on analytic theory, will be done by theorist elsewhere, and we will collaborate with them.

The feasibility of our flight experiment is based on our previous ground-based experiments, which confirm that synchronization of the DAW occurs. These ground-based experiments also revealed unexplained phenomena, including a nonharmonic synchronization state, which we will investigate with our flight data.

In the unlikely event that our main experiment with waves does not yield useful results, we have a plan for a backup experiment to investigate mobility-limited motion of particles having a different size.

BROADER IMPACTS

This research has an overlap with nonlinear dynamics as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, a website, and Twitter. A K-12 outreach presentation will be made at an elementary school.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

In this report period, our major accomplishment includes the publication of our first dedicated flight experiment on the International Space Station (ISS), advancing the understanding of nonlinear waves in three-dimensional dusty plasmas under microgravity. This flight experiment was performed in PK-4 Campaign 5 on the ISS, investigating a low-frequency compressional wave (called dust acoustic wave) self-excited in a collection of micron-size particles immersed in a neon plasma. These dust particles obtained large electric charges by absorbing electrons and ions in the plasma, constituting a dusty plasma (also called a complex plasma) in three dimensions. Our ground-based research effort included analyzing image data for dust motion recorded in our dedicated flight experiment and developing new data analysis method.
CORRELATION AND SPECTRUM OF DUST ACOUSTIC WAVES

Our research effort yielded the publication on the correlation and spectrum of dust acoustic waves (DAWs) under microgravity conditions. This result on DAWs was published in the journal Physics of Plasmas. Several highlights from this paper include: (1) We reported a steady and long-lived DAW, which had not been previously reported for PK-4. This wave was observed in our first PK-4 ISS flight experiment. (2) We introduced a correlation function for describing DAWs, to characterize the temporal decay of dust fluctuations. This decay was used as an indicator for the existence of any nonlinear processes that make density fluctuations coherent (instead of entirely random) or for the existence of unsteady phase conditions. (3) Our research approach was essentially microscopic, at atomic-like particle level. In obtaining the correlation and spectrum, we started from individual particles position, which was imaged using video cameras in our ISS flight experiment. This approach allows a direct comparison with statistical physics theories at the most fundamental levels.

Scientific knowledge we gained from this research includes:

• We observed the emergence of a broad spectrum of collective modes in a three-dimensional dusty plasma consisting of micron-size polymer microspheres confined in a glow-discharge plasma. In dusty plasmas like ours, DAWs are commonly expected to be excited in a narrow bandwidth due to the instabilities arising from ion flows. Our research showed that a spectrum of modes could be excited, over a broad range of wave numbers. This observation reveals that the fundamental physics mechanism for density fluctuations can arise from the presence of not only the instabilities, but also nonlinear coupling that could transfer energy from an unstable mode to other modes.

• We tested a previously-reported theoretical model for dust acoustic waves. Our experiment confirmed the predictions by the linear theory, regarding wave frequency varying with wave number (i.e., the so-called dispersion relation) and the most unstable modes. This test reveals the fundamental physics underlying DAWs include not only the effects due to charge separation, ion drift, electron-neutral collisions, ion-neutral collisions, dust-neutral collisions, but also kinetic

Task Progress:	effects of dust motion. • We confirmed nonlinear Cnoidal waveforms of dust acoustic waves in a plasma under microgravity conditions. In our previous ground-based experiment, we discovered Cnoidal waveforms (with sharpened peaks and fatten bottoms) in a dust cloud levitating in a plasma sheath. Our flight experiment confirmed this discovery in a dust cloud that was trapped in a plasma positive column. ANALYSIS OF ISS PK-4 FLIGHT DATA We developed a Bootstrap method in fitting the experimental dispersion relation for DAWs. Using this method, we were able to find not only the values for some fundamental physics quantities in our PK-4 dusty plasma (ion density, electron temperature, plasma electric field, dust charge and temperature, and so on), but also the uncertainties of these quantities. It is typically hard to extract these quantities from a microgravity experiment like ours, because the flight experiment facility is not equipped with some diagnostics that are typically available for ground-based experiment. We developed our own computer code for particle image velocimetry (PIV) method. This code was written in Matlab; it is used to evaluate the fluid velocity field in the dust cloud. A benefit from in-house developed software is that it can be easily tailored to our need more accurately. During this report period, we also continued our effort in finding signatures for wave synchronization and nonlinear wave interactions. This effort concentrated on studying physics phenomena such as partial synchronization and frequency clustering, synchronization states, the growth of nonlinear waves, three-wave mixing, wavefront merging, and so on. COLLABORATION WITH OTHER SCIENTISTS In preparing for our PK-4 paper published during the past year, we worked closely with PK-4 Core Team members in Germany and Russia as well as the PK-4 scientists at Unburn University. In general, our research requires close collaboration with several other institutions or agencies, including the PK-4 Core Team members, PK-4 Facility Science Team (FST), ESA (European Space Agency) officials, CADMOS (Microgravity and Spatial Operations Development Support Center) at the CNES (France National Center for Space Studies), and the collaborators specific to Iowa projects. THEORY During this period, we corresponded with theoretical collaborators in India (Prof. S. Tiwari, and Prof. Abhijit Sen) and Naval Research Laboratory (Dr. Guru Ganguli, and Dr. C. Crabtree). With these collaborators, we co-authored a paper on the theory is nonlinear wave mixing, under conditions related to our experiments.

Bibliography Type:	Description: (Last Updated: 01/23/2025)
Abstracts for Journals and Proceedings	Liu B, Goree J. "ISS flight experiments measuring acoustic waves and microscopic flows in three-dimensional dusty plasmas." Presented at the 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020
Abstracts for Journals and Proceedings	Liu B, Goree J, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Usachev AD, Molotkov VI, Petrov OF, Thoma MH. "Pulsed shear motion in a three-dimensional dusty plasma under microgravity conditions." Presented at the 62nd Annual Meeting of the APS Division of Plasma Physics, Virtual, November 9-13, 2020. Bulletin of the American Physical Society. 2020;65(11):Abstract: CP12.00004. http://meetings.aps.org/Meeting/DPP20/Session/CP12.4 , Nov-2020
Abstracts for Journals and Proceedings	Goree J, Liu B, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Usachev AD, Molotkov VI, Petrov OF, Thoma MH. "Correlation and spectrum of dust acoustic waves in a plasma under microgravity conditions." 62nd Annual Meeting of the APS Division of Plasma Physics, Virtual, November 9-13, 2020. Bulletin of the American Physical Society. 2020;65(11):Abstract: CP12.00005. http://meetings.aps.org/Meeting/DPP20/Session/CP12.5 , Nov-2020
Abstracts for Journals and Proceedings	Zhuravlyov V, Goree J. "Validity of moments of the dynamic structure factor in a dusty plasma." 62nd Annual Meeting of the APS Division of Plasma Physics, Virtual, November 9-13, 2020. Bulletin of the American Physical Society. 2020;65(11):Abstract: CP12.00012. http://meetings.aps.org/Meeting/DPP20/Session/CP12.12 , Nov-2020
Abstracts for Journals and Proceedings	Berumen J, Goree J. "Experimental measurement of viscoelasticity in a 2D dusty plasma using modulated shear flows." 62nd Annual Meeting of the APS Division of Plasma Physics, Virtual, November 9-13, 2020. Bulletin of the American Physical Society. 2020;65(11):Abstract: CP12.00013. http://meetings.aps.org/Meeting/DPP20/Session/CP12.13 , Nov-2020
Abstracts for Journals and Proceedings	Chaubey N, Goree J. "Dust charge reversal in an afterglow plasma." 62nd Annual Meeting of the APS Division of Plasma Physics, Virtual, November 9-13, 2020. Bulletin of the American Physical Society. 2020;65(11):Abstract: CP12.00018. http://meetings.aps.org/Meeting/DPP20/Session/CP12.18 , Nov-2020
Articles in Peer-reviewed Journals	Goree J, Liu B, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Molotkov VI, Usachev AD, Petrov OF, Thoma MH, Thomas E Jr, Konopka U, Prokopiev S. "Correlation and spectrum of dust acoustic waves in a radio-frequency plasma using PK-4 on the International Space Station." Phys Plasmas. 2020 Dec;27(12):123701. https://doi.org/10.1063/5.0024500 , Dec-2020
Articles in Peer-reviewed Journals	Mir A, Tiwari S, Goree J, Sen A, Crabtree C, Ganguli G. "A forced Korteweg–de Vries model for nonlinear mixing of oscillations in a dusty plasma." Phys Plasmas. 2020 Nov;27(11):113701. https://doi.org/10.1063/5.0022482 , Nov-2020
Articles in Peer-reviewed Journals	Kananovich A, Goree J. "Shocks propagate in a 2D dusty plasma with less attenuation than due to gas friction alone." Phys Plasmas. 2020 Nov;27(11):113704. https://doi.org/10.1063/5.0016504 , Nov-2020

Articles in Peer-reviewed Journals	Kananovich A, Goree J. "Experimental determination of shock speed versus exciter speed in a two-dimensional dusty plasma." Phys Rev E. 2020 Apr;101(4-1):043211. https://doi.org/10.1103/PhysRevE.101.043211 ; PMID: 32422787 , Apr-2020
Dissertations and Theses	Kananovich A. "Shock waves in dusty plasma." Dissertation, The University of Iowa, Iowa City, July 2020. , Jul-2020